

The Robotics Club of BRAC University (ROBU):

“Achieve your imagination through your robot”—the vision of the high-functioning sociopaths of ROBU! “

Here at BRAC university every club has a foundational meaning and purpose. With vibrant campus life the clubs are branches that breathe in new ideas, perspectives and bonds. Even among the clubs ROBU stands tall with its uniqueness in operations and accomplishments for a relatively young club of our country.

It operates on a foundational philosophy inspired by the late Sir Fazle Hasan Abed: "Small is beautiful, but big is necessary". Originating from the BRAC University Robotics LAB established in January 2012, the club has evolved from a small laboratory into a national launchpad for deep-tech innovation and global engineering competitiveness. ROBU was built on the ambitious legacy of projects like BRAC Onnesha, Bangladesh's first satellite, and the Bracu Chondrobot-2's entry into the NASA Lunabotics Mining Competition. With continuous legacies like these shining bright in the clubs portfolio the club prides itself in its curious and tenacious students. Now a pioneering robotics club in Bangladesh the club aims to spread the flag of our country even more prominently abroad.

We firmly believe in the power of curiosity and hard work. If you are enthusiastic about robotics and want to learn on the go this will be one of the most memorable journeys of your uni life. The club offers you a community of highly motivated individuals who are just as eager to share their insights into robotics as you are to learn. With productive workshops, competitions and mentoring offered at each turn there is never a dull moment in this club.

ROBU Functional Departments

Department	Primary Focus & Responsibilities
Strategic Planning (SP)	Shaping the long-term vision of the club, planning flagship national competitions (like TRACTION), and coordinating major robotics projects.
Finance & Marketing (F&M)	Securing sponsorships, managing project budgets (e.g., funding for rover and AUV teams), and promoting club initiatives.
Human Resources (HR)	Managing member recruitment, performance tracking, professional development workshops, and internal club culture.
Event Management	Executing logistics, venue setup, and operational flow for club-hosted events, seminars, and hackathons.
Arts & Design (A&D)	Creating all visual assets, event posters, UI/UX designs for projects, and maintaining the club's branding.
Information Technology (IT)	Managing the club's digital infrastructure, website, and technical workflows.
Editorial And Publication (E&P)	Combining editorial and outreach work alongside major publications and communication.

Organizational Hierarchy & Designations

ROBU follows a corporate-style progression track, allowing members to climb the ladder from entry-level apprentices to core panel leaders based on performance.

Tier	Designation	Role Description
Advisory	Faculty Advisor	A faculty member providing high-level academic, administrative, and technical guidance.
Core Panel	President	The executive head of the club, responsible for overall operations, external relations, and strategic direction.
Core Panel	Vice President	The second-in-command, overseeing internal department coordination and stepping in for the President when needed.
Core Panel	General Secretary	The operational anchor managing administrative duties, club records, and core communication.
Department Head	Director	The absolute head of a specific department (e.g., Director of Event Management), driving its goals and team output.
Sub-Panel	Assistant Director (AD)	Supports the Director in managing departmental workflows and team productivity.
Sub-Panel	Secretary	Manages the administrative and organizational tasks within a specific department.

Executive Body	Senior Executive	Experienced members who lead sub-teams or specific tasks within a department.
Executive Body	Executive	Established members handling core departmental operations and project execution.
Executive Body	Junior Executive	Members who have passed the apprentice phase and are taking on more autonomous responsibilities.
Entry Level	Apprentice (AP)	The starting designation for newly recruited members, focusing on learning the ropes and assisting the executives.

International Flagship Teams

ROBU members consistently bypass regional constraints to engineer machines capable of competing on the world's hardest stages.

- **BRACU Mongol-Tori:** The university's premier Mars rover team represents Bangladesh on global platforms organized by NASA and the Mars Society. The team recently achieved 7th place globally at the University Rover Challenge (URC) 2026, utilizing advanced ROS2 autonomy, differential GNSS RTK, and a custom 7-DOF robotic arm for extreme Martian analogs.
- **BRACU Duburi:** An autonomous underwater vehicle (AUV) team that pushes the boundaries of maritime robotics. The team recently secured 8th place among 58 international contenders at RoboSub 2025 and won a special "Entrepreneurship" award. They have directly qualified for the RoboSub 2025 Semi-Finals in Irvine, California, backed by their "Powered By" industry sponsor, Dubotech Inc..
- **BRACU ALTER and Dichari:** ROBU's autonomous drone and rescue rover initiatives. Dichari reached the European Robotics League (ERL) championship in Poland, finishing 4th and winning the Perseverance Award as the only Asian team. ALTER qualified for the RoboCup Rescue Robot League finals as the solitary South Asian representative among 17 global finalists.
https://www.youtube.com/watch?v=fEI0oUBJN_I
- **Team Automorph** at the World Invention Creativity Olympic 2026 held in Seoul, South Korea made the nation proud, securing Gold Medal and the prestigious Award of Excellence, placing us in the top 1% out of around 400 teams from 38+ countries! competed against powerhouse nations including the USA, China, Turkey, Iran, India, UK, Saudi Arabia, Mexico, Malaysia, New Zealand and many more.

National Triumphs & Hosted Events

- **World Robot Games 2026:** ROBU's "Team Stellar" emerged as the Champion in the Soccer Robot 1v1 category at the National Qualifiers, earning the honor to represent Bangladesh at the International Round in Chiba, Japan.
- **UIU CSE FEST 2025:** The club deployed four custom-built Soccer Bot teams to the tournament, demonstrating immense mechanical reliability and securing 4th place overall.
- **TRACTION (অভ্যুদয়):** ROBU's flagship, self-hosted national robotics competition. The massive event challenges engineers across multiple disciplines with high-stakes segments including RoboStrikers (soccer bots), MechaSprint (track racing), Pathfinder (precision line-following), Aqua Wars (surface water robots), Prompt Engineering, and the Innovators Exhibit.
- **ROBOSPHERE 2025** – The Intra-Departmental Robotics Competition, jointly hosted by the HR Department and IT Department of the Robotics Club of BRAC University - ROBU. The event brought three exciting segments: Line Follower Race (LFR), Prompt Engineering, and Project Showcase.

Hands-On Workshops & Skill Development ROBU actively bridges the gap between academic theory and industry readiness through rigorous, hands-on training sessions hosted by our internal departments. Some of our current and past projects include

- **Path to Automation (LFR Workshop):** Organized by the Arts & Design Department in February 2025, this hands-on workshop equips students with the technical skills to build and program smart Line-Following Robots.
- **Soccer Bot Workshop:** Hosted by the Event Management Department, this session brings in expert guest instructors to teach the mechanics and strategy behind building competitive 1v1 soccer bots.
- **LaTeX for Research & Resumes:** Conducted by former club leadership, this practical session utilizing Overleaf teaches members how to professionally format research articles, project documentation, and academic resumes.
- **GitHub & Career Optimization:** The club regularly hosts foundational bootcamps on GitHub version control alongside HR-led seminars featuring alumni guest speakers who teach resume writing and LinkedIn optimization
- **Embedded Hardware and Firmware Engineering Bootcamp:** This intensive program is designed to provide comprehensive, hands-on experience in industrial Embedded Systems Hardware and Firmware Engineering .Explore the intricacies of STM32 microcontrollers, advanced bit manipulation, and the complete hardware development lifecycle transforming foundational C/C++ knowledge into deployment-ready, commercial-grade architecture.

Landing page will have a logo of club and a group photo of members fading in the back
The theme will be focused on snappy dark but not too distracting
See what is popular and appropriate for a club intro/ recruitment and exposure intended for our purpose.

THE stack is github and render based with a structure and separation of files required for quick edits and debugs

Page 1

Start with intro of Club written in a clear striking font

Below will be our departments about us and join us button

Join us will have a embedded form link which shall open a new tab when clicked to google forms

Page 2

Loaded when clicked our departments shows in a neat stack all department full names.

Clicking on any department name will take you to another page with brief description and images of current club panel shown in a scrolling fashion

Page 3

The about us page will take you to menu which shows 2 options to choose from

Workshops and Milestones as stated in the document.

Each event will be described in brief with a drop down style showing only titles upon load
[description hidden]